

SCHOOL OF ENGINEERING, CUSAT
B. TECH DEGREE IInd SEMESTER FIRST INTERNAL EXAMINATIONS,
APRIL 2024

23-200-0201A COMPUTER PROGRAMMING AND PROBLEM SOLVING (ME-A)
(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

CO1	Elucidate the basic architecture and functionalities of a computer and also recognize the hardware parts.
CO2	Apply programming constructs of C language to solve real-world problems.
CO3	Explore user-defined data structures like arrays in implementing solutions to problems like searching and sorting
CO4	Explore user-defined data structures like structures, unions and pointers in implementing solutions
CO5	

PART A

(Answer all questions)

(5 x 2= 10 Marks)

No.	Questions	Marks	BL	CO	PO
I (a)	Draw a flow chart to check whether a given number is even number or not.	2	L1	1	1-5
(b)	Compare and contrast Compiler and Interpreter	2	L2	1	1-5
(c)	What will be the output of the following C code? <pre>#include <stdio.h> void main() { int x = 4, y, z; y = --x; z = x--; printf("%d%d%d", x, y, z); }</pre>	2	L3	2	1-5
(d)	Using conditional operator check whether the given number is divisible by 11	2	L3	2	1-5
(e)	Differentiate between while and do while with an example.	2	L2	2	1-5

PART B
(Answer the following questions
choosing either the first OR second option)
(1 x 10= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Explain the components of a Digital Computer System with a neat architecture	6	L1	1	1-5
(b)	Write an algorithm to swap (exchange) the values of two variables using temporary variable.	4	L2	1	1-5
OR					
III (a)	Explain the different types of programming languages?	6	L1	1	1-5
(b)	Write an algorithm for printing Fibonacci series.	4	L2	1	1-5
(1 x 10= 10 Marks)					
IV (a)	Write a c program to check whether a given number is palindrome or not	5	L3	2	1-5
(b)	Is there any difference between the pre-decrement and post decrement operators? Explain with suitable examples.	5	L1	2	1-5
OR					
V (a)	What is a token? What are different types of tokens available in C language? Explain.	5	L1	2	1-5
(b)	Differentiate between break and continue with an example.	5	L2	2	1-5

Bloom's Taxonomy Levels

L1 - %	L2- %	L3-%	L4-%	L5-%	L6-%
48	34	18	-	-	-